

Simple But Iconic: Implications for Designing Icons Representing Delays in Time-Critical Medical Events

Katherine A. Zellner
Drexel University, College for
Computing and Informatics
United States
kz376@drexel.edu

Vidhi Shah
Drexel University, College for
Computing and Informatics
United States
vps43@drexel.edu

Aleksandra Sarcevic
Drexel University, College for
Computing and Informatics
United States
as3653@drexel.edu

ABSTRACT

Decision-support tools that increase awareness of time-critical pending tasks can help reduce delays during life-saving interventions. However, conveying the urgency of tasks through visual alerts remains a challenge. This case study describes our design and evaluation of delay alerts represented by icons within a decision-support system for emergency medical events such as trauma resuscitation. We used iterative sketching to design a series of icons for three delay alerts and two online surveys with a total of 55 clinicians to evaluate the icons. Although we have prioritized interpretable icons based on common icon design principles, clinicians preferred the simplest rather than the most interpretable icons. This case study contributes design guidelines for icons representing complex processes and concepts that can be applied to other types of time-critical teamwork.

CCS CONCEPTS

• **Human-centered computing** → Visualization; Visualization design and evaluation methods.

KEYWORDS

Alerts, Icons, Icon design, Clinical Decision-Support System (CDSS), Survey, Complex workflows, Emergency medicine, Trauma resuscitation

ACM Reference Format:

Katherine A. Zellner, Vidhi Shah, and Aleksandra Sarcevic. 2024. Simple But Iconic: Implications for Designing Icons Representing Delays in Time-Critical Medical Events. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3613905.3637137>

1 INTRODUCTION

Traumatic injuries are a significant cause of death for individuals under 45 in the United States, with 50% of preventable deaths linked to delays or errors in the initial evaluation and management of severely injured patients [28]. With each 3-minute delay in care

for severely injured adults with hypotension, the risk of death increases by 1% [8]. To mitigate delays, prior interventions have focused on improving access to required tools like refrigerators with blood products [24] or thawed plasma [17]. These tool-based interventions have decreased some process delays but have not addressed skill-based decision-making errors, the most prominent disruptors in emergency care workflows [18]. A new approach that can detect and alert clinicians to delays could facilitate timely decisions, addressing this gap and reducing errors and delays. Our long-term research goal is to develop a decision-support system that detects delays in the performance of life-saving interventions during trauma resuscitation and then alerts the trauma team to these delays through an information display. Appropriate alert design is critical to our research because clinicians are concerned that alerts can cause distraction and hinder situational awareness [10]. Our prior work found that clinicians preferred visual alerts that communicated ongoing delays at a glance [29]. These results suggest that well-designed icons could represent delays and associated tasks.

Icons are a popular design element because they are easily recognizable and memorable [1, 23] and more globally recognized than text [14]. Icons have also been found to provide better cognitive support [6] and improve the usability of graphical user interfaces compared to text-based forms or labels [19]. Despite user preferences for icons over text-based interfaces [16], poorly designed or researched icons can result in ambiguous interpretations, leading to ineffective systems [3, 12]. To create more transparent and meaningful icons, researchers have studied visual icon design, focusing on designing medical icons [20] and icons for children [4], the classification of icons [11], and the arrangement of icons within specific interfaces [15, 22]. In the icon design literature, icons are evaluated on five metrics: *detectability* (the ability to locate an icon among others), *legibility* (the ability to determine what an icon depicts), *interpretability* (the ability to match an icon with its intended meaning), *recognizability* (the ability to remember the meaning of an icon over time), and *preference* (the greater liking of one option over others) [23, 26, 27]. Previous research on icon design has found that the most realistic and easiest-to-interpret icons have better outcomes for untrained and trained users [5, 7, 15]. Users generally prefer icons that they rate as most realistic and that have the highest interpretability.

In emergency medical settings, prior HCI research has designed icons for an electronic medical record [20]. These icons represent simple concepts such as medical orders, family trees, and patient history. In addition, the nature of interaction with an electronic medical record lends more affordances to users, including the search

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3637137>



Figure 1: *Left:* A mock-up of the prototype alert system in the resuscitation room (left wall monitor) and the ambient LED lights that light up with a triggered alert (room ceiling). *Right:* An enlarged view of the prototype alert system display.

function and the ability to hover over an icon for a description [20]. In this case study, we also aimed to design icons for a new system that alerts medical teams to delays in the performance of life-saving interventions. Our goal, however, was to design icons that convey complex concepts that do not have pre-existing designs, such as incomplete clinical tasks and delays in multi-step procedures. Icon design for complex concepts has been reported for only a few domains, including online privacy regulation [9]. Because our icons are displayed on a wall monitor, they must be designed for quick reference and use without direct interaction. To achieve these goals, we used a multi-step design and evaluation approach. First, we conducted an initial sketching exercise to develop icons representing the most common delays in trauma resuscitation. We then surveyed clinicians at our research site to elicit feedback on the initial icon designs. We revised the icons and then administered another survey based on this feedback. Using this approach and expert feedback, we derived three guidelines for designing icons for dynamic and time-critical medical teamwork: (1) prioritize simplicity, (2) implement text and symbols with care, and (3) design and evaluate icons within the context of the system and its environment.

2 BACKGROUND, RESEARCH SETTING, AND PARTICIPANTS

Prior research on trauma teamwork identified two significant challenges in maintaining temporal awareness during resuscitations [13]: (1) providers have difficulty in perceiving time and remembering when activities occurred, and (2) providers have difficulty in conveying the urgency of tasks. As part of our prior work, we developed a new prototype alert system to support delay awareness during trauma resuscitation based on interviews, design workshops, and near-live simulations with clinicians (Figure 1) [29]. The system focuses on the three most common delays observed by clinicians during resuscitations: (1) “No IV/IO Access”—missing intravenous (IV) or intraosseous (IO) access needed for administering fluids or

blood products; (2) “No Vital Signs or Blood Pressure”—missing patient vitals or blood pressure from the monitor, and (3) “Incomplete Request”—an incomplete physician request for action such as blood transfusion, intubation or medication [29]. The system uses one of two 38” x 22” large wall monitors in the room and displays a grid with different alerts (Figure 1). The central area of the grid shows the elapsed time, while the outer segments display icons representing different delays. When a new alert is triggered, the corresponding icon appears and lights up in one of the grid areas (Figure 1, right). The grid design allowed for multiple alerts to be shown simultaneously, as multiple delays could co-occur. For example, a team leader may request an additional IV access point to administer fluids or medications. If the team does not successfully establish the additional access point after five minutes, the icon would appear on the screen. Once the delay has been resolved, the icon will fade away. For clinicians who cannot see the monitor because of their role-specific tasks and positions, we developed a concept of lights around the ceiling that light up with each grid alert (Figure 1, left).

This case study describes the design and evaluation process of our new icons for the prototype alert system. We conducted the study at a level-one pediatric trauma center in an urban pediatric hospital. Trauma resuscitations at our research site are performed by a multidisciplinary medical team involving surgical fellows and residents, emergency medicine physicians, anesthesiologists, respiratory therapists, bedside nurses, medication nurses, and social workers. Each team member performs a series of tasks, collectively evaluating and managing the injured patient using the Advanced Trauma Life Support protocol [25].

We invited 310 clinicians to participate in our surveys via email listservs, receiving responses from 27 and 28 clinicians for the first and second surveys, respectively. The first survey participants included 13 emergency medicine attending physicians, five registered nurses, three emergency medicine fellows, three physicians

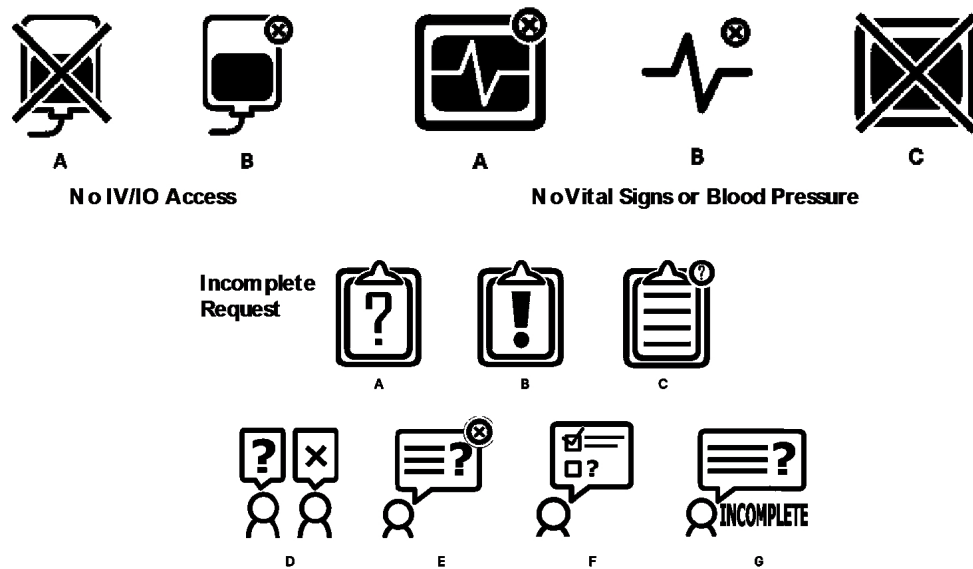


Figure 2: Initial icon designs for the “No IV/IO Access,” “No Vital Signs or Blood Pressure,” and “Incomplete Request” delay alerts.

(non-specified), one nurse practitioner, one psychiatric program coordinator, and one pharmacist. The second survey participants included 11 emergency medicine attending physicians, three emergency medicine fellows, 12 registered nurses, one physician assistant, and one physician (non-specified). On average, participants in the first survey had nine years of experience in trauma resuscitation, and participants in the second survey had 13 years of experience. Survey participants were eligible for a \$30 Amazon gift card draw upon completing the survey.

3 THE INITIAL ICON DESIGN PROCESS FOR DELAY ALERTS

We began the icon design process by using iterative sketching. A graphic designer on our team used the 10x10 sketching technique [2], producing ten different high-level icon ideas for each of the three delay alerts and then ten more for each iteration. We discussed each design idea and compiled preferred features into five icon designs for each alert type, leading to 15 initial designs. We shared these 15 designs with the medical experts on our research team and then re-sketched based on their feedback. After discussing the designs with the research team, we created high-fidelity graphics of three to seven icons per delay alert and evaluated them using a design survey (Figure 2).

The initial sketches for the “No IV/IO Access” alert began as various depictions of an IV fluid bag. We added an “X” sign to the designs to convey the idea of the IV/IO’s absence. We used an EKG line to represent vital signs to design the “No Vital Signs or Blood Pressure” alert. One variation featured an EKG line with an “X” sign, another showed a monitor with an “EKG” label, and the third showed only a monitor and an “X” sign. The “Incomplete Request” icons were among the most challenging to design because they required conveying the most abstract alert concept. The initial icon

designs featured speech bubbles with question marks or incomplete items on a checklist. As we further discussed the icons with medical experts on our team, we created seven designs for this alert: three were variations based on a checklist item visual, and four were variations based on speech bubbles with question marks.

4 FIRST SURVEY: METHODS AND FINDINGS

We conducted a 10-minute online survey using Qualtrics to assess user preference and interpretability of the initial icon designs (Figure 2). The survey was piloted by three medical experts for clarity and length and included four sections. It began with questions about participant roles and experience, followed by evaluation questions for 12 icons representing three delay alerts. Participants interpreted each icon, ranked them based on preference, and provided qualitative feedback on the alert grid and ambient lights.

The results from the first survey showed that the “No IV/IO Access” alert icons were the most interpretable, with 40% (11/27) of participants correctly interpreting Icon A and 34% correctly interpreting Icon B (Table 1). Sixty percent of participants preferred Icon B over Icon A because Icon A had a large “X” sign across the fluid bag. Participants thought that the “X” sign obfuscated the meaning of the alert. One participant wrote, “[This] looks like stop the IV fluids to me.” Another participant described the icon as “a clinical instruction to NOT to give blood.”

The “No Vital Signs or Blood Pressure” alert was the next most interpretable, with 17% correctly interpreting Icon A and 37% choosing Icon B as the best representation of the delay. Six out of 27 participants commented that icon designs for this alert were too simplistic, including “It just looked like a big black X. The vitals part is unclear.”

Finally, the icons for the “Incomplete Request” alert were the least interpretable, with 14% (4/27) correctly identifying Icon D

Table 1: Percentage of participants who correctly interpreted and preferred each icon in the first survey. Icon letters correspond to the icons from Figure 2.

Delay Alert	Icon Version	Correctly Interpreted	Preferred
"No IV/IO Access"	A	40%	40%
	B	34%	60%
"No Vital Signs or Blood Pressure"	A	10%	37%
	B	13%	37%
	C	17%	20%
"Incomplete request"	A	4%	0%
	B	4%	43%
	C	4%	20%
	D	14%	0%
	E	4%	43%
	F	0%	0%
	G	0%	20%

and even fewer correctly interpreting all other icon options. Icons B and E were most preferred, with 43% of participants choosing these over others. Participants did not feel that the icons offered sufficient information. One participant wrote, "*The speech bubbles don't make sense to me.*" Additionally, 60% of the participants indicated confusion about the alert in their open-ended responses and suggested breaking it down into the most common incomplete requests with specific labels referring to incomplete tasks.

Although the icon design literature has shown that the high interpretability of icons is linked with high preferability [7, 15], our findings indicated that interpretability may not be the most critical factor for icon design in a time-critical medical setting. The feedback from the first survey suggested that participants valued icons with clean designs and fewer visual details more than icons with high interpretability.

5 ICON REDESIGN FOR DELAY ALERTS

Using the findings and participant feedback from the first survey, we redesigned the initial icons (Figure 3). We removed the "X" symbol from all designs to reduce the complexity and added new elements to improve the interpretability of icons. For example, a needle and graded lines were added to the "No IV/IO Access" alert, and elements representing heart rate and blood pressure were included in the "No Vital Signs or Blood Pressure" alert. We also created multiple graphical representations to convey the three most frequent incomplete requests in trauma resuscitation: medication administration, blood transfusion, and intubation (inserting a tube into the patient's airway to support breathing). These graphical representations formed the basis for three new variants of the "Incomplete Request" alert: Incomplete Medicine Request, Incomplete Blood Request, and Incomplete Intubation Request. For each of these alerts, we created a graphic version and a labeled progress bar version. Through this process, we prioritized clean designs based on user preferences from the first survey. We discussed the redesigned icons with medical experts on our team before finalizing them for the second survey.

Based on our experiences from the first survey, we were interested in finding out if the improved designs would again show the

lack of association between high interpretability and high preferability. We also wanted to know if adding text to the icons improved their recognizability and likeability.

6 SECOND SURVEY: METHODS AND FINDINGS

The second 10-minute survey, also on Qualtrics, compared the redesigned icons with the most preferred and recognizable icons from the first survey. The second survey had two sections. In the first section, we measured the interpretability and legibility of the redesigned icons by asking participants to describe what they thought the icon represented. The second section asked participants to rank icons in the order of preference using three scales from the icon evaluation literature [7, 15]: most preferred to least preferred, simplest to most complex, and most realistic to most abstract. For the preferability ranking, participants also had the option of each icon with or without added text. For redesigns of the "Incomplete Request" alert, participants were asked to rank only the redesigned versions with the progress bar against the new graphical icon versions. We did not ask them to rank the initial design with speech bubbles because it represented a different concept and was not an appropriate comparison. At the end of each ranking, participants could provide the rationale behind their decisions in a free-text field. We did not assess icon detectability because our prototype alert system displays only a limited number of icons, reducing the need for a user to differentiate between icons like on a phone [1].

All redesigned icons had increased interpretability scores after implementing feedback from the first survey (Table 2). The redesigned icon E for the "No IV/IO Access" alert was correctly interpreted by 82% (23/28) of participants, compared to 40% who correctly interpreted the initial design. The new icon E for the "No Vital Signs or Blood Pressure" alert was correctly interpreted by 60% of participants, compared to 10% of participants for the initial design. The new designs for the "Incomplete Request" alert and icons for medication administration, blood transfusion, and intubation also had high interpretability scores, 78%, 78%, and 70%, respectively.

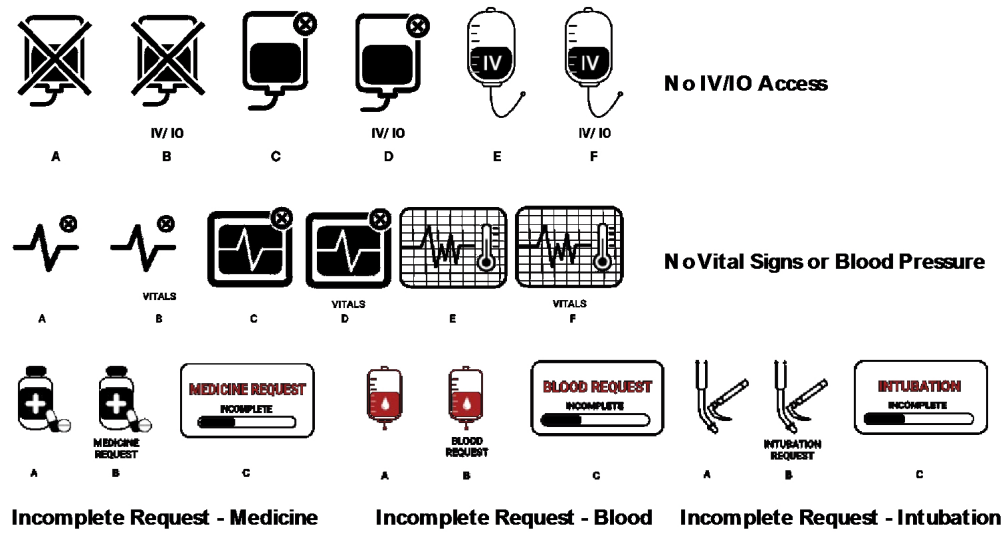


Figure 3: The redesigned icons evaluated in the second survey for each delay alert.

However, we observed low preferability scores for new redesigns across almost all icon types (Table 3). Of the 28 participants, only 1% ranked the redesigned icon A for the “No IV/IO Access” alert as their preferred icon. Fifteen percent of participants ranked the redesigned icon E for the “No Vital Signs or Blood Pressure” alert as their preferred vitals icon, and 22% ranked the graphical icons for medication administration, blood transfusion, and intubation as their preferred icons. Participants preferred the redesigned icon F with the added IV/IO text for “No IV/IO Access” more than without text (18% vs. 11%, respectively). Of the 28 participants, 74% rated the redesigned icon E for the “No Vital Signs or Blood Pressure” as the most complex, noting that these icons contained too many elements. Their comments included: “The icon is too graphically busy” and “I had to think about what the wiggly line meant plus the thermometer – it made it more complex.” The added details did not contribute to more clarity for the users. One participant described, “The simplest [icon] is also the most realistic and sufficient.”

The redesigned icon E for the “No IV/IO Access” alert was rated the simplest (Figure 3). However, five of the 28 participants stated they preferred the icons with the “X” symbol because it took an additional cognitive step to understand that IV/IO access is delayed or missing without the “X.” Adding text within the icon seemed to improve the interpretability of the design. One participant wrote, “I did not even realize that this graphic meant IV access until you have put IV/IO under it. Without it, I thought it meant IVF.” Although the added text “IV” within the redesigned icon did not provide enough clarity, the entire label “IV/IO” did.

The survey participants did not prefer the icon versions with added text for the “Incomplete Request” alert. Instead, they preferred the labeled progress bar icon because it was “more straightforward” than the one with graphics alone. Even so, ten participants noted that they preferred the labeled progress bar but admitted it could be confusing if the only change between the three alerts was just in text. One participant explained, “Pictures are generally easier

to interpret quickly than words.” Despite their preference for the graphic icons, participants required more clarity, as one explained: “Just the pill bottle is insufficient.” Although the graphical representation of intubation had a relatively high interpretability score, participants noted that it “took them a while” or was “a very confusing image.” Four participants described how the more complex icons, like that for intubation, added to their cognitive load and required too much time attempting to decipher their meaning.

Most redesigned icons received similar “abstract” and “realistic” scores (Table 2), indicating that most participants did not see a significant difference in the level of abstraction between the icons. For example, 33% of participants found the redesigned icon E for the “No Vital Signs or Blood Pressure” alert to be the most abstract. In contrast, 37% of participants found this icon to be the most realistic. The only icon with a noticeable difference in “abstract” vs. “realistic” scores was the redesigned icon E for the “No IV/IO Access” alert, with 52% of participants rating it as the most realistic and just 15% of participants rating it as the most abstract. Our participants did not refer to the abstraction or realism of the icons in their qualitative feedback but instead focused on whether the icons were “complicated” or “confusing.” Our data shows preferability is most associated with simplicity rather than interpretability. Like our first survey, the findings from the second survey showed that icons with the highest interpretability were not the most preferred icons.

7 DISCUSSION

In this case study, we designed and evaluated a series of icons for a prototype alert system that represent the most common delays in the performance of life-saving interventions during trauma resuscitation. Unlike prior studies on icon design in emergency medical settings [20], we focused on designing icons that communicate complex concepts and processes. Our icons needed to convey delays in establishing IV/IO access, obtaining vital signs, and completing critical tasks before life-saving procedures could proceed. Our icons

Table 2: Percentage of participants who correctly interpreted, preferred, and ranked icons without added text labels on three scales in the second survey. Icon letters correspond to the icons from Figure 3.

Delay Alert	Icon	Correctly Interpreted	Preferred	Ranked Complex	Ranked Simplest	Ranked Realistic	Ranked Abstract
"No IV/IO Access"	A	40%	1%	33%	26%	15%	44%
	C	34%	15%	18%	18%	18%	26%
	E	82%	11%	37%	44%	52%	15%
"No Vital Signs or Blood Pressure"	A	10%	28%	11%	26%	33%	15%
	C	17%	15%	4%	59%	15%	41%
	E	60%	15%	74%	8%	37%	33%
"Incomplete Request – Medicine"	A	78%	22%	56%	44%	30%	70%
	C	-	77%	60%	40%	44%	56%
"Incomplete Request – Blood"	A	78%	22%	78%	22%	26%	74%
	C	-	77%	56%	44%	44%	56%
"Incomplete Request – Intubation"	A	70%	22%	56%	44%	29%	71%
	C	-	37%	44%	56%	48%	52%

also differed from those described in prior work because they were designed for an informative user interface that is quickly glanced upon, without any interactions or affordances for users to learn about the alert.

In our design process, we initially focused on easily interpretable icons because prior icon design literature found that users generally prefer icons with the highest interpretability [7, 15]. However, as we found in the first survey and confirmed in the second, clinicians at our research site primarily valued icon simplicity, even when the icons were challenging to interpret. Our participants better understood the redesigned and more complex icons in the second survey but still preferred the simpler, initial icons from the first survey. For example, almost all participants correctly interpreted the redesigned icon representing vital signs but preferred the more straightforward, less correctly interpreted icon from the initial set of icons. We found similar results with the "Incomplete Request" icon redesign, where participants generally preferred the labeled progress bar and rated it as a simpler option. As we observed in open-ended user feedback, every additional design element added to clinicians' cognitive load, which is already increased in high-stress clinical scenarios. For example, adding text or symbols like an "X" sign to provide specificity or convey that something is missing (e.g., "No IV/IO Access") also added confusion and extended interpretation times. Instead of reading the "X" symbol as "IV/IO access missing or not established," our participants thought the symbol meant "do not administer fluids." Similarly, adding text underneath the graphic representation had variable effects on the participants. In the "No IV/IO Access" icon, the addition of the label "IV/IO" confused several participants, leading them to interpret it as "fluids." Adding the entire label clarified the "No IV/IO Access" and "No Vital Signs or Blood Pressure" alerts.

7.1 Design Guidelines: Icons for Complex Concepts in Time-Critical Teamwork

Our findings suggest that clinicians working in a high-stress, time-critical medical environment may have different priorities than users studied in previous icon design research, where the most realistic and easiest-to-interpret icons were superior to other icons [7, 15]. As we progressed through the icon design process, three guidelines for designing icons for dynamic, time-critical teamwork emerged: (1) prioritize simplicity, (2) implement text and symbols with care, and (3) design and evaluate icons within the context of the system and its environment.

Prioritize simplicity: From this case study, we learned that icons for high-stress, time-critical teamwork require simple designs. Comments from our participants such as, "I had to think about what the wiggly line meant plus the thermometer" and "Pictures are generally easier to interpret quickly than words," suggested that interpreting each icon took additional time and effort. Unnecessary and extra elements like speech bubbles or background texture can add confusion rather than clarity. In time-critical medical teamwork such as trauma resuscitation and emergency medicine, information overload can lead to fatigue and impaired decision making [21]. Simplicity in icon design facilitates quick recognition and decision-making efficiency. Several strategies can be used to prioritize simplicity in icon design. First, the design could limit the number of elements in an icon, focusing only on what is needed to convey the concept. Second, the design should avoid ambiguity introduced through non-standard graphical representations or symbols. Although symbols like an "X" may seem universal, our case study showed they can confuse users and obfuscate the intended meaning. To determine which elements to include or exclude, we found that iterative user testing elicited clear and constructive feedback, allowing us to refine icons and identify the essential elements for each design.

Table 3: Percentage of participants who preferred icons with added text vs icons without added text in the second survey. Icon letters correspond to the icons from Figure 3.

Delay Alert	Icon	Preferred
“No IV/IO Access”	A	1%
	B	0%
	C	15%
	D	18%
	E	11%
	F	18%
“No Vital Signs or Blood Pressure”	A	15%
	B	26%
	C	15%
	D	0%
	E	15%
	F	18%
“Incomplete Request – Medicine”	A	22%
	B	11%
“Incomplete Request – Blood”	A	22%
	B	11%
“Incomplete Request – Intubation”	A	22%
	B	19%

Implement text and symbols with care: Although text can enhance user understanding, its length and its clarity must be carefully considered to avoid hindering icon interpretation. Concise and well-chosen text can complement symbols, providing necessary context without overwhelming the icon’s visual simplicity. For example, in the “No IV/IO Access” icon participants preferred the addition of the “IV/IO” label under the icon but did not find the “IV” within the icon helpful. Additionally, designers should consider environmental aspects when integrating text, ensuring it is universally comprehensible and inclusive across user groups. Striking the right balance between text and symbols is essential for creating effective and intuitive icons.

Design and evaluate icons within the context of the system and its environment: Designers should consider the context of the system when conceptualizing the icons. The number of icons needed to convey delays in a time-critical setting differs from those needed for electronic medical records or home screens on mobile applications. Users may respond differently if they have less time to interpret each icon or if they must evaluate the icons within the context of the system interface. The environmental context, such as time constraints, cognitive load, and the pace of decision-making, should also be taken into consideration throughout the design process. Although we briefly described our prototype alert system at the beginning of each survey, our icons were evaluated outside the system context and without time pressure, which might have limited our findings. In our future work, we will evaluate the icons within the context of work and with time pressure using real-world simulation.

In addition to evaluating our icons outside the system interface and work context, our results must be considered against two other

study limitations. First, our study was conducted at a single research site, which could limit the generalizability of our findings. The cultures, practices, and workflows may vary across different institutions, potentially influencing the preferences and interpretations of the icon designs. Collaborating with multiple research sites or diverse healthcare facilities could offer a more comprehensive understanding of icon preferences in varying contexts. Second, the methodology of using surveys to evaluate icons has inherent limitations, particularly in the context of visual and dynamic elements. Although previous research used surveys to evaluate icons in general contexts, this approach may be more limited for time-critical work and complex medical concepts. Instead, a multi-method strategy combining interviews or focus groups and surveys could ensure a more nuanced understanding of the icon designs and their effectiveness, usability, and preferability. Surveys typically use static images and collect responses in a linear manner, which may not fully capture icon interpretation and the nuanced and dynamic nature of their use in actual work. Icons convey information through visual cues, and their effectiveness relies on immediate and intuitive comprehension. In future work, we plan to evaluate the overarching system in live simulations to evaluate the effectiveness of the computerized support and the icon design.

8 CONCLUSION

In this case study, we described the design and evaluation of three delay alerts represented by icons within a decision-support system for emergency medical events such as trauma resuscitation. We evaluated the icon designs by conducting two surveys with a total of 55 clinicians. Through this process, we successfully designed

interpretable icons representing complex concepts, such as missing intravenous (IV) and intraosseous (IO) access and incomplete requests for life-saving procedures. We found that clinicians preferred the simplest rather than the most interpretable icons. Based on these results, we identified three guidelines for designing icons that represent complex concepts and processes in dynamic, time-critical teamwork: (1) prioritize simplicity, (2) implement text and symbols with care, and (3) design and evaluate icons within the context of the system and its environment.

ACKNOWLEDGMENTS

We thank the staff at Children's National Hospital for participating in this research. This research was funded by the National Science Foundation under the research grant number IIS-1763509. We also thank our research collaborators, Randall S. Burd, MD, PhD, Travis Sullivan, MD, Ryan Dixon, and Waverly Gestrich-Thompson.

REFERENCES

- [1] Matthias Böhmer and Gernot Bauer. 2010. Exploiting the icon arrangement on mobile devices as information source for context-awareness. In *Proceedings of the 12th international conference on Human computer interaction with mobile devices and services (MobileHCI '10)*. Association for Computing Machinery, New York, NY, USA, 195–198. <https://doi.org/10.1145/1851600.1851633>
- [2] Bill Buxton *Sketching user experience: Getting the design right and the right design*. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA, 2007. <https://doi.org/10.1016/B978-0-12-374037-3.X5043-3>
- [3] Hongin Cheng and Patrick E. Patterson. 2007. Iconic hyperlinks on e-commerce websites. *Appl Ergon*, 38, 1 (Jan 2007), 65–69. <https://doi.org/10.1016/j.apergo.2006.01.007>
- [4] Shuhui Chiu, Chornng-Shiuh Koong and Sa-Hui Fan. 2012. Icon design principles for preschoolers: Implications derived from child development. *Procedia - Social and Behavioral Sciences*, 64 (2012), 228–237. <https://doi.org/10.1016/j.sbspro.2012.11.027>
- [5] Ho Chun-Heng and Hou Kai-Chun. 2015. Exploring the attractive factors of app icons. *KSII Trans Internet and Info Sys*, 9, 6 (Jun 2015). <https://doi.org/10.3837/tiis.2015.06.016>
- [6] William W. Gaver. 1991. Technology affordances. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '91)*. Association for Computing Machinery, New York, NY, USA, 79–84. <https://doi.org/10.1145/108844.108856>
- [7] Ravindra S. Goonetilleke, Heloisa Martins Shih, Hung Kai On and Julien Fritsch. 2001. Effects of training and representational characteristics in icon design. *Int J Hum Comput Stud*, 55, 5 (2001), 741–760. <https://doi.org/10.1006/ijhc.2001.0501>
- [8] Russel L. Gruen and Gregory Jurkovich. 2006. Patterns of errors contributing to trauma mortality: Lessons learned from 2,594 deaths. *Ann Surg*, 244, 3 (Sep 2006), 371–380. <https://doi.org/10.1097/01.sla.0000234655.83517.56>
- [9] Hana Habib, Yixin Zou, Yaxing Yao, Alessandro Acquisti, Lorrie Cranor, Joel Reidenberg, Norman Sadeh, and Florian Schaub. 2021. Toggles, dollar signs, and triangles: How to (In)effectively convey privacy choices with icons and link texts. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems (CHI '21)*. Association for Computing Machinery, New York, NY, USA, Article 63, 1–25. <https://doi.org/10.1145/3411764.3445387>
- [10] Yue-Yung Hu, Alexander F. Arriaga, Emilie M. Roth, Sarah E. Peyre, K. A. Corso, R. S. Swanson, R. T. Osteen, P. Schmitt, A. M. Bader, M. J. Zinner and C. C. Greenberg. 2012. Protecting patients from an unsafe system: The etiology and recovery of intraoperative deviations in care. *Ann Surg*, 256, 2 (Aug 2012), 203–210. <https://doi.org/10.1097/SLA.0b013e3182602564>
- [11] Sarah J. Isherwood, Siné J. P. McDougall and Martin B. Curry. 2007. Icon identification in context: The changing role of icon characteristics with user experience. *Hum Factors*, 49, 3 (Jun 2007), 465–476. <https://doi.org/10.1518/001872007X200102>
- [12] Andre W. Kushniruk and Vimla L. Patel. 2004. Cognitive and usability engineering methods for the evaluation of clinical information systems. *J Biomed Inform*, 37, 1 (Feb 2004), 56–76. <https://doi.org/10.1016/j.jbi.2004.01.003>
- [13] Diana S. Kusonoki and Aleksandra Sarcevic. 2015. Designing for temporal awareness: The role of temporality in time-critical medical teamwork. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)*. Association for Computing Machinery, New York, NY, USA, 1465–1476. <https://doi.org/10.1145/2675133.2675279>
- [14] Kenneth N. Loddington. 1983. Iconic interfacing. *IEEE Computer Graphics and Applications*, 3, 2(1983), 11–20. <https://doi.org/10.1109/MCG.1983.262982>
- [15] Carlos Nakamura and Qing Zeng-Treitter. 2012. A taxonomy of representation strategies in iconic communication. *Int J Hum Comput Stud*, 70, 8 (Aug 1 2012), 535–551. <https://doi.org/10.1016/j.ijhcs.2012.02.009>
- [16] Jakob Nielsen. 2007. Miniatures versus icons as a visual cache for videotex browsing. *Behaviour & Information Technology*, 9, 6 (2007), 441–449. <https://doi.org/10.1080/01449299008924258>
- [17] Deborah J. Novak, Yu Bai, Rhonda K. Cooke, Marisa B. Marques, Magali J. Fontaine, Jerome L. Gottschall, et al. 2015. Making thawed universal donor plasma available rapidly for massively bleeding trauma patients: experience from the Pragmatic, Randomized Optimal Platelets and Plasma Ratios (PROPPR) trial. *Transfusion* 55, 6 (2015), 1331–1339. <https://doi.org/10.1111/trf.13098>
- [18] Sarah E. Henrickson Parker, Aaron A. Laviana, Rishi K. Wadhera, Douglas A. Wiegmann, and Thoralf M. Sundt. 2010. Development and evaluation of an observational tool for assessing surgical flow disruptions and their impact on surgical performance. *World J Surg* 34, 2 (2010), 353–361. <https://doi.org/10.1007/s00268-009-0312-z>
- [19] Yucel Batu Salman, Hongin Cheng, Ji Young Kim and Patrick E. Patterson. 2010. Medical information system with iconic user interfaces. *International Journal of Digital Content Technology and its Applications*, 4, 1 (2010), 137–148. <https://doi.org/10.4156/jdcta.vol4.issue1.14>
- [20] Yusel B. Salman, Hongin Cheng and Patric Patterson. 2012. Icon and user interface design for emergency medical information systems: A case study. *Int J Med Inform*, 81, 1 (Jan 2012), 29–35. <https://doi.org/10.1016/j.ijmedinf.2011.08.005>
- [21] Laura Saffi, James Walton, John Blenkinsopp and Graham Walton. 2020. Information overload in emergency medicine physicians: A multisite case study exploring the causes, impact, and solutions in four North England National Health Service Trusts. *Journal of Medical Internet Research* 22, 7 (Jul. 2020), e19126. <https://doi.org/doi:10.2196/19126>
- [22] Johanna M. Silvennoinen and Jussi P. P. Jokinen. Aesthetic appeal and visual usability in four icon design eras. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems (CHI '16)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/2858036.2858462>
- [23] Paul F. Sorenson and Jayson M. Webb. 1991. Experimental evaluation of icon quality. *ACM SIGCHI Bulletin*, 23, 4 (Oct 1991). <https://doi.org/10.1145/126729.1056076>
- [24] Travis M. Sullivan, Zachary P. Mileston, P. Emily C. Alberto, Elizabeth M. Waibel, Waverly V. Gestrich-Thompson, Peyton E. Tempel, Jennifer I. Fritzeen, Karen J. O'Connell, Jacob Cheng, and Randall S. Burd. 2022. Reducing the Time to Blood Administration after Pediatric Injury: A Quality Improvement Initiative. *Pediatric Quality and Safety* 7, 3 (2022) 563 DOI: <https://doi.org/10.1097/pq9.0000000000000563>
- [25] American College of Surgeons' Committee on Trauma. The ATLS Subcommittee and The International ATLS Working Group. 2013. Advanced trauma life support (ATLS): The ninth edition. *J Trauma and Acute Care Surg*, 74, 5 (2013), 1363–1366.
- [26] Jayson M. Webb, Paul F. Sorenson and Neal P. Lyons. 1989. An empirical approach to the evaluation of icons. *ACM SIGCHI Bulletin*, 21, 1 (1989), 87–90. <https://dl.acm.org/doi/10.1145/67880.67894>
- [27] Susan Wiedenbeck. 1999. The use of icons and labels in an end user application program: An empirical study of learning and retention. *Behaviour & Information Technology*, 18, 2 (1999), 68–82. <https://doi.org/10.1080/01449299919129>
- [28] Jiaquan Xu, Sherry Murphy, Kenneth Kochanek and Elizabeth Arias. 2021. Mortality in the United States, *NCHS Data Brief*. 2022 Dec;(456):1–8.
- [29] Katherine A. Zellner, Aleksandra Sarcevic, Megan A. Krensta, Travis M. Sullivan, and Randall S. Burd. 2023. Understanding delay awareness and mitigation mechanisms through an iterative design and evaluation of a prototype alert system for complex teamwork. *Proc. ACM Hum.-Comput. Interact*, 7, CSCW2, Article 260 (October 2023), 28 pages. <https://doi.org/10.1145/3610051>